

RAKESH R J

Associate Software Engineer | .NET Developer | Computer Vision Engineer

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PROFESSIONAL SUMMARY

Software engineer with approximately 2.7 years of total professional experience spanning .NET development, Azure cloud services, REST API integration, SQL data operations, and computer vision engineering. Seeking opportunities to leverage enterprise application development, cloud, and AI skills in a challenging and growth-oriented environment while delivering impactful solutions.

WORK EXPERIENCE

Accenture

Jan 2025 – Present | ~1.9 Years

Associate Software Engineer (ASE)

Project: Roche Diagnostics – Production Application and Data Support

Enterprise production support involving .NET, Azure, SQL, and REST API-based data workflows.

- Handled and resolved production tickets related to application, data, and processing issues.
- Analyzed tickets and investigated root causes to identify the required resolution.
- Performed database changes and data updates using SQL based on business requirements.
- Used REST APIs and Swagger/OpenAPI to trigger and process data in the production environment.
- Uploaded and managed required data using Azure Storage Accounts.
- Triggered downstream data processing workflows and pipelines through Azure.
- Monitored and validated data processing results after API and Azure operations.
- Troubleshoot issues involving .NET applications, databases, APIs, and data-processing workflows.
- Followed production change and incident-management processes while working with live enterprise systems.
- Collaborated with the team to resolve production tickets and complete requested changes efficiently.

Technologies: .NET | C# | SQL | Azure | Azure Storage | REST API | Swagger/OpenAPI | Azure Data Pipelines | Production Support

Computer Vision Engineer

Project 1: PPE and Safety Alerting System

AI-based safety monitoring system for PPE violations and forklift-related incidents.

- Developed Python-based components for processing computer vision detection results.
- Used gRPC for inter-service communication between microservices.
- Built alert-processing workflows for PPE and forklift use cases.
- Used MongoDB to manage alert configurations, jobs, channels, and related data.
- Integrated AWS SQS for asynchronous alert processing and AWS Lambda-based notification workflows.
- Integrated notification services including AWS SES and SNS for real-time alerting.

Technologies: Python | gRPC | MongoDB | AWS SQS | AWS Lambda | AWS SES | AWS SNS

Project 2: Gemini AI Video Analysis and Alerting

AI-powered video analysis using Gemini for safety event detection (helmet violations, falls, harness).

- Developed and modified FastAPI backend services for video ingestion and processing.
- Integrated Gemini AI for video analysis and generated structured detection results.
- Created and maintained prompts for detection scenarios including helmet violations, falls, and safety harness.
- Processed videos stored in Amazon S3 and optimized frame sampling rates for performance.
- Integrated detection results with an alerting pipeline via AWS SQS and stored data in MongoDB.

Technologies: Python | FastAPI | Gemini AI | Prompt Engineering | AWS S3 | AWS SQS | MongoDB | REST API

Project 3: Vehicle Detection and Number Plate Recognition

Computer vision system for vehicle detection, tracking, classification, and ANPR.

- Implemented vehicle detection using YOLOv8 with object tracking across video frames.
- Performed number plate detection and text extraction using EasyOCR.
- Classified vehicles into categories: cars, buses, trucks, bikes, ambulances, and auto-rickshaws.
- Implemented vehicle color classification and generated structured output including Track ID, type, plate, confidence, and color.
- Optimized video processing using frame skipping and frame-similarity techniques.

Technologies: Python | YOLOv8 | OpenCV | EasyOCR | TensorFlow | Keras | Object Tracking | Computer Vision

TECHNICAL SKILLS

Programming and Data: Python, SQL, C#, .NET, Java, C

.NET Development: .NET, C#, ASP.NET, Production Support, Troubleshooting, Debugging

Cloud and DevOps: Microsoft Azure, Azure Storage Accounts, Azure Data Pipelines, Git, GitHub, CI/CD fundamentals

APIs and Integration: REST APIs, Swagger, OpenAPI, API Testing, Data Integration

AI and Computer Vision: Python, OpenCV, YOLOv8, EasyOCR, FastAPI, Gemini AI, TensorFlow, Keras, CNN, Object Tracking, Prompt Engineering

Cloud (AWS): AWS SQS, AWS Lambda, AWS SES, AWS SNS, Amazon S3

Databases and Tools: SQL, MongoDB, gRPC, VS Code, IntelliJ, Jupyter Notebook, Power BI, Git

Web: HTML, CSS, JavaScript, React.js, PHP

EDUCATION

Kongu Engineering College

April 2024

B.E. Computer Science Engineering — CGPA: 8.45 / 10.0

Veveaham Hr. Sec. School

March 2020

HSC (Class XII) — 70% | SSLC (Class X) — 84.8%